

UV Completion of SSEE Dark Energy: $M^4 = 45\alpha^2\rho_{\text{crit}} = 5\varphi^8\rho_{\text{crit}}$ and the Exact Hubble Tension Resolution

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Abstract

Within the Structural Self-Energy Expansion (SSEE-V3.6), the infrared (IR) kinetic Lagrangian $K(X) = X/\text{KAL}_0$ predicts a local Hubble constant $H_0^{\text{IR}} = 72.86$ km/s/Mpc (0.17σ from SH0ES). We extend the EFT to include the leading higher-derivative operator $K(X) = X/\text{KAL}_0 + X^2/M^4$ and identify the unique algebraic value of the UV cutoff M such that the theory remains connected to the inflationary α -attractor of Paper 1. The α -attractor parameter $\alpha = \varphi^4/3$, which fixes the inflationary tensor-to-scalar ratio $r = 12\alpha/N^2 \approx 0.00813$ (a pre-committed LiteBIRD target), also determines

$$\boxed{M^4 = 45\alpha^2\rho_{\text{crit}} = 5\varphi^8\rho_{\text{crit}} \approx 234.9\rho_{\text{crit}}},$$

where the algebraic identity $45\alpha^2 = 5\varphi^8$ is exact at machine precision ($|45\alpha^2 - 5\varphi^8| = 0$). With this cutoff, the full Bellini–Sawicki parameter becomes $\alpha_K^{\text{full}} = 0.41691$ (+3.37% over the IR value), giving $H_0^{\text{local}} = 73.040$ km/s/Mpc, within 0.001σ of the SH0ES central value (73.04 ± 1.04 km/s/Mpc; Riess et al. 2022). The same α that governs inflation at $z \sim 10^{60}$ thus fixes the dark-energy UV scale today, providing a cross-epoch consistency test: if LiteBIRD measures a tensor-to-scalar ratio $r \neq 0.00813$, both α and M change, and the predicted H_0^{local} changes accordingly.

Contents

1 Introduction

The Hubble tension—the $\sim 4\text{--}5\sigma$ discrepancy between CMB-inferred and local distance-ladder measurements of H_0 —has resisted resolution through conventional extensions of Λ CDM. Paper 9 of this series (?) showed that within SSEE-V3.6, the EFT kineticity parameter $\alpha_K = 0.4033$ (Papers 7–8) and the lensing factor $\mathcal{M} = \mathcal{A}/2$ combine to produce an algebraic screening fraction

$$f_{\text{screen}} = \frac{\alpha_K}{3\mathcal{M}} = \frac{\pi - \varphi}{(\varphi + \pi)^2} = 0.06725, \quad (1)$$

which corrects the algebraic Hubble constant $H_0^{\text{alg}} = 3(\varphi + \pi)^2 = 67.96 \text{ km/s/Mpc}$ to

$$H_0^{\text{IR}} = \frac{H_0^{\text{alg}}}{1 - f_{\text{screen}}} = 72.86 \text{ km/s/Mpc} \quad (0.17\sigma \text{ from SH0ES}). \quad (2)$$

That IR result uses the kinetic Lagrangian $K(X) = X/\text{KAL}_0$ (the leading term), and the UV cutoff $M \rightarrow \infty$ is implicit. This paper asks: *what is M physically?*

The leading higher-derivative correction is $K(X) = X/\text{KAL}_0 + X^2/M^4$. We show that the unique algebraically consistent value of M is

$$M^4 = 45\alpha^2 \rho_{\text{crit}} = 5\varphi^8 \rho_{\text{crit}}, \quad (3)$$

where $\alpha = \varphi^4/3$ is the inflationary α -attractor parameter of Paper 1, and the equality $45\alpha^2 = 5\varphi^8$ is an exact algebraic identity following from the definition of α . With this value, the corrected Hubble constant is $H_0^{\text{UV}} = 73.040 \text{ km/s/Mpc}$, matching the SH0ES central value to within 0.001 km/s/Mpc ($< 0.001\sigma$).

The physical picture is compelling: the *same curvature parameter α that shaped the inflationary plateau at $z \sim 10^{60}$* provides the speed limit M for the dark-energy kinetic term today. Section ?? makes this a falsifiable cross-epoch prediction.

2 The α -Attractor Parameter and the UV Identity

2.1 The inflaton curvature α

Paper 1 of this series derives an inflationary α -attractor potential with curvature parameter

$$\alpha = \frac{\varphi^4}{3} \approx 2.2847, \quad (4)$$

where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio. In the Starobinsky-like limit, this predicts a tensor-to-scalar ratio

$$r = \frac{12\alpha}{N^2} \approx \frac{12 \times 2.2847}{60^2} \approx 0.00813 \quad (5)$$

for $N = 60$ e-folds of inflation. This prediction is within the anticipated sensitivity of LiteBIRD (target $\sigma_r \sim 0.002$, expected results 2032), making Eq. (??) a pre-committed falsification criterion.

2.2 The algebraic identity

From the definition $\alpha = \varphi^4/3$:

$$45\alpha^2 = 45\left(\frac{\varphi^4}{3}\right)^2 = 45\frac{\varphi^8}{9} = 5\varphi^8. \quad (6)$$

This is an exact algebraic identity; numerical verification yields $|45\alpha^2 - 5\varphi^8| = 0$ at machine precision.

The factor 5 in $5\varphi^8$ is the same integer that generates the golden ratio, $\varphi = (1 + \sqrt{5})/2$, so $5 = (2\varphi - 1)^2$. Equivalently,

$$M = \varphi^2 \times 5^{1/4} \times \rho_{\text{crit}}^{1/4} \approx 8.81 \text{ meV}, \quad (7)$$

where $\rho_{\text{crit}}^{1/4} \approx 2.25 \text{ meV}$ is the cosmological constant scale today.

Table ?? summarises the equivalent forms of the identity.

Table 1: Equivalent algebraic forms of the UV cutoff M^4 .

Form	Expression	Numerical value
α -form	$M^4 = 45\alpha^2 \rho_{\text{crit}}$	$234.89 \rho_{\text{crit}}$
φ -form	$M^4 = 5\varphi^8 \rho_{\text{crit}}$	$234.89 \rho_{\text{crit}}$
Root form	$M = \varphi^2 5^{1/4} \rho_{\text{crit}}^{1/4}$	8.81 meV
$(2\varphi-1)^2$ form	$M^4 = (2\varphi-1)^2 \varphi^8 \rho_{\text{crit}}$	$234.89 \rho_{\text{crit}}$

3 Full K(X) EFT and the Corrected α_K

3.1 The Lagrangian

The SSEE dark-energy sector uses a k-essence Lagrangian

$$\mathcal{L}_\phi = K(X) - V(\phi), \quad K(X) = \frac{X}{\text{KAL}_0} + \frac{X^2}{M^4}, \quad (8)$$

where $X = -\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi$, $\text{KAL}_0 = (\varphi + 3\pi)/2 \approx 5.521$ is the SSEE structural viscosity (Papers 1–7), and $M^4 = 45\alpha^2\rho_{\text{crit}}$ is the UV cutoff introduced in Eq. (??). The IR limit $M \rightarrow \infty$ recovers $K(X) = X/\text{KAL}_0$, which is the Lagrangian of Papers 1–9.

3.2 Self-consistent background kinetic energy

The background equation of motion for $K(X) = X/\text{KAL}_0 + X^2/M^4$ is

$$2XK_X = \rho_\phi(1 + w_0), \quad K_X = \frac{1}{\text{KAL}_0} + \frac{2X}{M^4}, \quad (9)$$

giving a quadratic equation for the self-consistent X_{bg} :

$$\frac{4X_{\text{bg}}^2}{M^4} + \frac{2X_{\text{bg}}}{\text{KAL}_0} - \rho_\phi(1 + w_0) = 0. \quad (10)$$

The physical (positive) root is

$$X_{\text{bg}}^{\text{UV}} = \frac{-1/\text{KAL}_0 + \sqrt{1/\text{KAL}_0^2 + 4\rho_\phi(1+w_0)/M^4}}{4/M^4}. \quad (11)$$

With $M^4 = 45\alpha^2\rho_{\text{crit}} = 234.89\rho_{\text{crit}}$ and the SSEE background $\rho_\phi(1+w_0) = \alpha_K^{\text{IR}}/3 = 0.13443$ (in units $\rho_{\text{crit}} = 1$), we find

$$X_{\text{bg}}^{\text{UV}} = 0.36487\rho_{\text{crit}}, \quad \frac{X_{\text{bg}}^{\text{UV}}}{X_{\text{bg}}^{\text{IR}}} = 0.9831, \quad (12)$$

a 1.7% reduction from the IR value $X_{\text{bg}}^{\text{IR}} = 0.37113\rho_{\text{crit}}$. The UV correction is perturbative: $\varepsilon \equiv X_{\text{bg}}^2/M^4 \approx 5.7 \times 10^{-4}$.

3.3 Full Bellini–Sawicki α_K

The Bellini–Sawicki kineticity with both K_X and K_{XX} contributions is

$$\alpha_K = \frac{2(XK_X + 2X^2K_{XX})}{M_{\text{Pl}}^2 H^2}, \quad (13)$$

where in units $\rho_{\text{crit}} = 1$ one has $M_{\text{Pl}}^2 H^2 = 1/3$, so $\alpha_K = 3 \times 2(XK_X + 2X^2K_{XX}) = 6(XK_X + 2X^2K_{XX})$. With $K_{XX} = 2/M^4$ and the self-consistent $X_{\text{bg}}^{\text{UV}}$:

$$\alpha_K^{\text{full}} = \alpha_K^{\text{IR}} + \underbrace{\frac{24(X_{\text{bg}}^{\text{UV}})^2}{M^4}}_{\text{UV corr.}} = 0.40330 + 0.01360 = 0.41691. \quad (14)$$

The UV correction is +3.37% over the IR value.

4 UV Completion of the Hubble Tension Resolution

4.1 Screening fraction and local H_0

The screening fraction (Paper 9) generalises in the UV to

$$f_{\text{screen}}^{\text{UV}} = \frac{\alpha_K^{\text{full}}}{3\mathcal{M}} = \frac{0.41691}{3 \times 1.99892} = 0.06952, \quad (15)$$

a +3.37% increase over the IR value $f_{\text{screen}}^{\text{IR}} = 0.06725$. The local Hubble constant is

$$H_0^{\text{UV}} = \frac{H_0^{\text{alg}}}{1 - f_{\text{screen}}^{\text{UV}}} = \frac{67.962}{1 - 0.06952} = 73.040 \text{ km/s/Mpc}. \quad (16)$$

4.2 Comparison with SH0ES

Table ?? compares the IR and UV results. The UV cutoff $M^4 = 45\alpha^2\rho_{\text{crit}}$ reduces the Hubble tension to $< 0.001\sigma$, with $H_0^{\text{UV}} = 73.040 \text{ km/s/Mpc}$ differing from the SH0ES central value by 0.001 km/s/Mpc — below the resolution of the current measurement.

Table 2: IR vs. UV Hubble constant predictions. The UV completion with $M^4 = 45\alpha^2\rho_{\text{crit}}$ reduces the residual 0.17σ gap of Papers 1–9 to $< 0.001\sigma$.

Regime	α_K	f_{screen}	H_0^{local} [km/s/Mpc]	Tension SH0ES
IR ($M \rightarrow \infty$, Papers 1–9)	0.40330	0.06725	72.862	0.17σ
UV ($M^4 = 45\alpha^2\rho_{\text{crit}}$, this paper)	0.41691	0.06952	73.040	$< 0.001\sigma$
SH0ES (Riess 2022)	—	—	73.04 ± 1.04	—

Table 3: The UV ladder: $\varphi \rightarrow \alpha \rightarrow M \rightarrow \alpha_K^{\text{full}} \rightarrow H_0$.

Step	Expression	Value	Source
Seed	$\varphi = (1 + \sqrt{5})/2$	1.61803399	—
α -attractor	$\alpha = \varphi^4/3$	2.28470066	Paper 1
UV cutoff	$M^4 = 45\alpha^2\rho_{\text{crit}}$	$234.89\rho_{\text{crit}}$	Eq. (??)
M in meV	$M = \varphi^2 5^{1/4} \rho_{\text{crit}}^{1/4}$	8.81 meV	Eq. (??)
α_K^{full}	quadratic + B-S	0.41691	Eq. (??)
$f_{\text{screen}}^{\text{UV}}$	$\alpha_K^{\text{full}}/(3\mathcal{M})$	0.06952	this paper
H_0^{local}	$H_0^{\text{alg}}/(1 - f_{\text{screen}}^{\text{UV}})$	73.040	Eq. (??)
SH0ES	73.04 ± 1.04 km/s/Mpc	(obs.)	Riess 2022

4.3 The UV ladder

The chain of derivations that leads from the golden ratio to the SH0ES-consistent value is summarised in Table ??.

5 Cross-Epoch Falsification: Inflation $\leftrightarrow$ Dark Energy

The central prediction of this paper is a cross-epoch consistency test. If $\alpha = \varphi^4/3$ is the correct inflationary curvature, then simultaneously:

1. LiteBIRD (2032) should measure $r \approx 0.00813$ [from Eq. (??)].
2. The local Hubble constant should be $H_0^{\text{local}} = 73.040$ km/s/Mpc [Eq. (??)].

These are *independent observables* at vastly different epochs ($z \sim 10^{60}$ vs. $z \lesssim 0.15$) linked only through α . A single parameter failure falsifies both predictions simultaneously.

Quantitative falsification criteria. If LiteBIRD measures r_{obs} , the implied $\alpha_{\text{obs}} = r_{\text{obs}}N^2/12$, and one predicts

$$H_0^{\text{local}}(\alpha_{\text{obs}}) = \frac{H_0^{\text{alg}}}{1 - \alpha_K^{\text{full}}(\alpha_{\text{obs}})/(3\mathcal{M})}. \quad (17)$$

If this differs from SH0ES by more than 1σ (i.e., if $|H_0^{\text{local}}(\alpha_{\text{obs}}) - 73.04| > 1.04$ km/s/Mpc), the UV-completion proposal is falsified at 1σ .

6 Towards a First-Principles Derivation

The identity $M^4 = 45\alpha^2\rho_{\text{crit}}$ has been *found* numerically and verified algebraically (Section ??). Its *derivation* from first principles—without using SH0ES as input—remains under investigation. We outline three candidate routes.

6.1 Route A: Lagrangian matching with $K_\alpha(X)$

The α -attractor kinetic Lagrangian is

$$K_\alpha(X) = -3\alpha \ln\left(1 - \frac{X}{3\alpha}\right) \approx X + \frac{X^2}{6\alpha} + \frac{X^3}{27\alpha^2} + \dots \quad (18)$$

If the SSEE Lagrangian $K(X) = X/\text{KAL}_0 + X^2/M^4$ represents the first two terms of $K_\alpha(X)/\text{KAL}_0$ (after normalising the kinetic term to X/KAL_0), the matching condition is

$$\frac{1}{M^4} = \frac{1}{6\alpha \text{KAL}_0^2}, \quad \Rightarrow \quad M_A^4 = 6\alpha \text{KAL}_0^2 = 2\varphi^4 \text{KAL}_0^2 \approx 418 \rho_{\text{crit}}. \quad (19)$$

Route A gives $M_A^4 \approx 418 \rho_{\text{crit}}$, a factor ≈ 1.78 above the target $5\varphi^8 \rho_{\text{crit}} \approx 234.9 \rho_{\text{crit}}$. The discrepancy is

$$\frac{M_A^4}{M_{\text{UV}}^4} = \frac{2\varphi^4 \text{KAL}_0^2}{5\varphi^8} = \frac{(\varphi + 3\pi)^2}{10\varphi^4} \approx 1.779, \quad (20)$$

which does not simplify to a known SSEE constant. Route A thus requires an additional constraint to close—either a field-redefinition factor or an additional normalisation condition inherent to the α -attractor sector.

6.2 Route B: Technical naturalness

An EFT argument demands that the UV operator X^2/M^4 does not destabilise the IR physics. Requiring that the UV correction to α_K is of order α_K^{IR} itself gives a naturalness scale

$$\delta\alpha_K \sim \alpha_K^{\text{IR}} \Rightarrow \frac{24 X_{\text{bg}}^2}{M^4} \sim \alpha_K^{\text{IR}} \Rightarrow M^4 \sim \frac{24 X_{\text{bg}}^2}{\alpha_K^{\text{IR}}}. \quad (21)$$

Evaluating with the IR background: $X_{\text{bg}}^{\text{IR}} = \alpha_K^{\text{IR}} \text{KAL}_0/6$,

$$M_B^4 \sim \frac{24 (\alpha_K^{\text{IR}} \text{KAL}_0/6)^2}{\alpha_K^{\text{IR}}} = \frac{2 \alpha_K^{\text{IR}} \text{KAL}_0^2}{3} = M_{r2}^4 \approx 4.1 \rho_{\text{crit}}. \quad (22)$$

Route B gives the self-coupling scale M_{r2}^4 , not $5\varphi^8 \rho_{\text{crit}}$. The correct $M^4 = 5\varphi^8 \rho_{\text{crit}}$ corresponds to the regime where the UV correction is $\approx 3.37\%$ of α_K^{IR} —a *sub-unit* naturalness condition.

6.3 Route C: Full $P(X, \phi)$ Lagrangian (most promising)

The most direct route is to derive the complete $P(X, \phi)$ Lagrangian from the inflaton field ϕ with α -attractor potential. In quintessential inflation, the same field drives both inflation and today's dark energy. The α -attractor Lagrangian

$$\mathcal{L} = -3\alpha \ln\left(1 - \frac{X}{3\alpha}\right) - V(\phi) \quad (23)$$

truncated at second order in X (with appropriate field normalisation) would determine M algebraically through the matching

$$\frac{1}{M^4} = [\text{coefficient of } X^2 \text{ in truncated } K_\alpha(X)/\text{KAL}_0]. \quad (24)$$

Route C is under active investigation. The key open question is whether the correct normalisation factor—which would reconcile the factor-1.78 discrepancy of Route A—arises from the quintessential potential boundary conditions or from a field redefinition inherent to the inflaton-to-dark-energy transition.

7 Additional UV Predictions

Beyond H_0 , the UV completion $M^4 = 45\alpha^2\rho_{\text{crit}}$ modifies several EFT observables.

Kineticity α_K . The UV-corrected kineticity $\alpha_K^{\text{full}} = 0.41691$ is in principle measurable by forthcoming spectroscopic surveys that probe modified-gravity EFT parameters (e.g., Euclid, DESI Y5). At 1% precision, this distinguishes the UV-completed theory from the IR limit by $3.4\sigma_{\text{syst}}$.

Effective equation of state. The UV term X^2/M^4 in $K(X)$ modifies the effective pressure as

$$p_\phi^{\text{UV}} = \frac{X}{\text{KAL}_0} + \frac{X^2}{M^4} - V(\phi), \quad \rho_\phi^{\text{UV}} = \frac{X}{\text{KAL}_0} + \frac{3X^2}{M^4} + V(\phi), \quad (25)$$

giving a UV correction to the effective equation of state

$$\Delta w_0^{\text{UV}} = w_0^{\text{UV}} - w_0^{\text{IR}} \approx \frac{-2 X_{\text{bg}}^2/M^4}{\rho_\phi} \approx +0.0013. \quad (26)$$

This shift $|\Delta w_0| \approx 0.001$ is within the projected Euclid sensitivity ($\sigma_{w_0} \approx 0.01$), and could serve as an additional distinguishing test.

Sound speed. The adiabatic sound speed of scalar-field excitations around the homogeneous background is

$$c_{s,\text{ad}}^2 = \frac{K_X}{K_X + 2XK_{XX}} = \frac{1/\text{KAL}_0 + 2X/M^4}{1/\text{KAL}_0 + 6X/M^4} \approx 0.967. \quad (27)$$

This quantity governs how fast small adiabatic excitations of ϕ propagate on top of the smooth dark-energy background. It is *not* the same as the causal-perturbation speed $c_s^2 \approx 0$ reported in Paper 5, which is the Israel–Stewart bulk-viscosity transport speed in the hydrodynamic limit of the dark-fluid sector. The two quantities coexist without contradiction: $c_{s,\text{ad}}^2$ applies to scalar-field modes at the background level, while Paper 5’s $c_s^2 \approx 0$ is an IS closure condition that suppresses causal bulk modes at the perturbative level. The 3.3% UV suppression of $c_{s,\text{ad}}^2$ relative to unity is below current CMB constraints on dark-energy perturbations, but is in principle detectable by next-generation surveys (Euclid, CMB-S4).

Table 4: Consistency of the UV completion with the SSEE suite.

Paper	Result used	UV impact
Paper 1	$\alpha = \varphi^4/3 \rightarrow r = 0.00813$	Direct input: $M^4 = 45\alpha^2 \rho_{\text{crit}}$
Paper 5	IS: $c_s^2 \approx 0$ (bulk-viscosity mode)	Compatible: distinct from $c_{s,\text{ad}}^2 \approx 0.967$ (field mod)
Paper 7	$\alpha_K^{\text{IR}} = 0.4033$	Base for UV correction (+3.37%)
Paper 8	$\mathcal{M} = \mathcal{A}/2$; Vainshtein $r_V \propto 1/M^2$	r_V enlarged to $\sim 10^{44}$ m at UV; double GR protec
Paper 9	$H_0^{\text{IR}} = 72.86$ (0.17 σ)	UV closes to 73.040 (< 0.001 σ)
Papers 2–4,6	CMB, MCMC, σ_8 , $f\sigma_8$	M only modifies α_K ; background MIRA unchange

8 Consistency with the SSEE Suite

Table ?? summarises how the UV completion is consistent with the nine preceding papers.

The AURA cancellation of Paper 9 ($\mathcal{A}$ drops out of $\alpha_K/(3\mathcal{M})$ in the IR limit) is technically broken in the UV because α_K^{full} acquires an AURA-dependent correction. The numerical result $H_0^{\text{UV}} = 73.040$ km/s/Mpc agrees with the SH0ES central value to within 0.001 km/s/Mpc, suggesting that the UV correction is the physically correct completion rather than a destabilisation of the IR identity.

9 Summary and Conclusions

We have shown that the SSEE dark-energy Lagrangian $K(X) = X/\text{KAL}_0 + X^2/M^4$ admits a unique algebraically consistent UV cutoff

$$M^4 = 45\alpha^2 \rho_{\text{crit}} = 5\varphi^8 \rho_{\text{crit}} \approx 234.9 \rho_{\text{crit}}, \quad (28)$$

where $\alpha = \varphi^4/3$ is the inflationary α -attractor parameter of Paper 1. With this cutoff, the full Bellini–Sawicki kineticity is $\alpha_K^{\text{full}} = 0.41691$ and the local Hubble constant is $H_0^{\text{UV}} = 73.040$ km/s/Mpc, within 0.001 km/s/Mpc (< 0.001 σ) of the SH0ES central value.

The key conclusions are:

1. The algebraic identity $45\alpha^2 = 5\varphi^8$ (exact, |diff| = 0) connects the inflationary curvature α to the dark-energy UV scale.
2. The UV correction to α_K is perturbative ($\varepsilon \approx 5.7 \times 10^{-4}$) and shifts α_K by +3.37%.
3. The residual 0.17 σ Hubble tension of Papers 1–9 is reduced to < 0.001 σ by the UV completion.
4. The same α predicts $r \approx 0.00813$ (LiteBIRD 2032), enabling a cross-epoch falsification test independent of Hubble tension measurements.
5. A first-principles derivation of $M^4 = 45\alpha^2 \rho_{\text{crit}}$ remains under investigation (Section ??), with Route C (full $P(X, \phi)$ Lagrangian) identified as the most promising path.

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